

ML
Unsupervised Dimensionality Reduction
(PCA)

Unsupervised Learning

- **Definition** (Wikipedia)

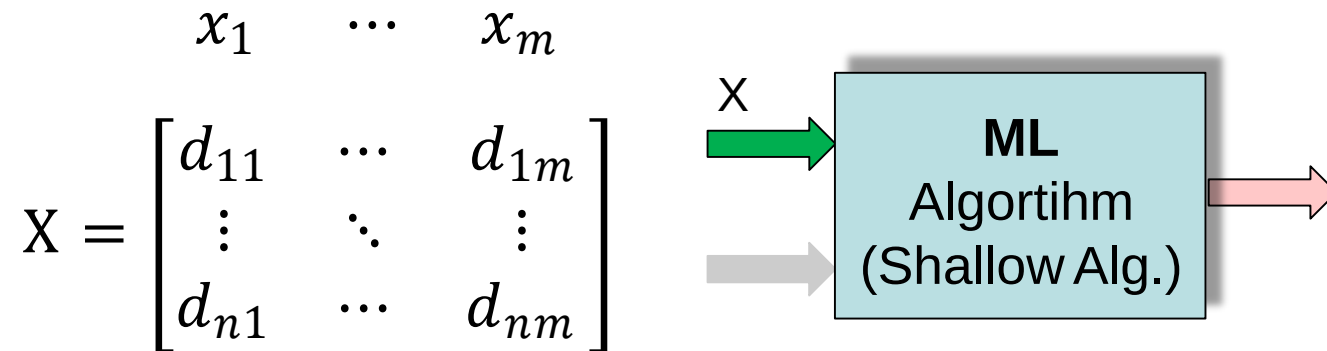
Unsupervised learning is a type of machine learning that looks for previously undetected patterns in a data set with no pre-existing labels and with a minimum of human supervision.

Machine Learning

- Supervised learning {
 - Linear Regression
 - Logistic Regression
 - Unsupervised {
 - Clustering
 - Market basket
 - Dimensionality Reduction
 - Reinforcement & Semi-supervised
- } - PCA

Principal Component Analysis (PCA)

Principal Component Analysis

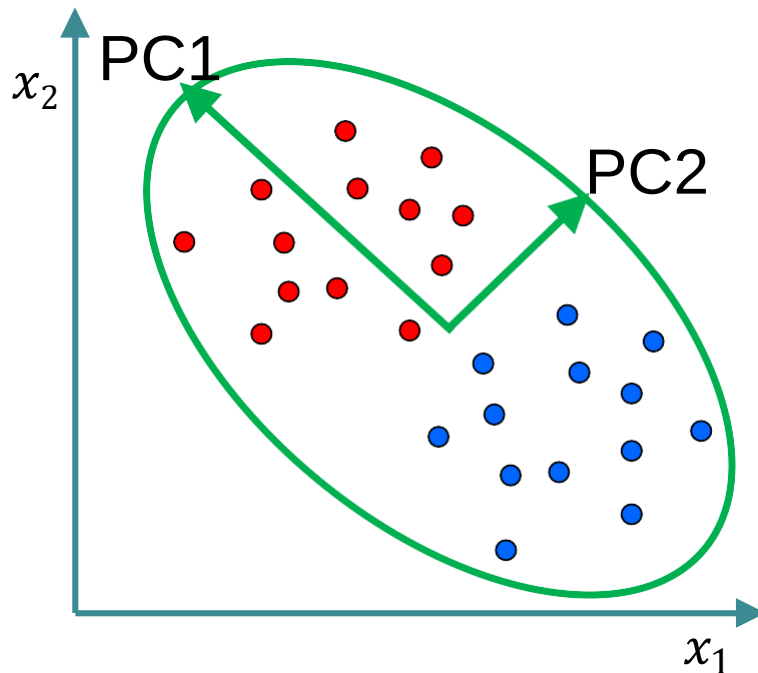


PCA

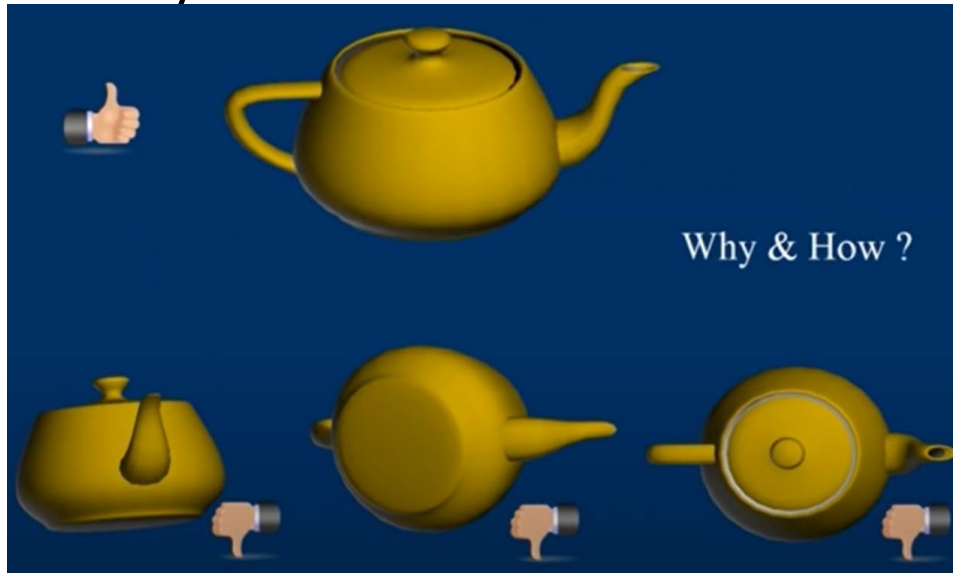
From m -independent variables of a dataset, PCA extracts the $p \leq m$ new independent variables that explain most of the dataset variance, regardless of the value of the dependent variable (unsupervised algorithm).

Principal Component Analysis

- 1° Scale the variables of the characteristic matrix (X)
- 2° Calculate the covariance matrix of the independent variables of X
- 3° Calculate the values and eigenvectors (vectores propios o autovectores) of the covariance matrix.
- 4° Choose a percentage P of explained variance and choose the largest $p \leq m$ eigenvalues such that
$$\frac{\sum_{j=1}^p \lambda_{jj}}{\sum_{i=1}^p \lambda_{ii}} \geq P$$
- 5° The eigenvectors associated with these larger p-values are the main components.



Principal Component Analysis



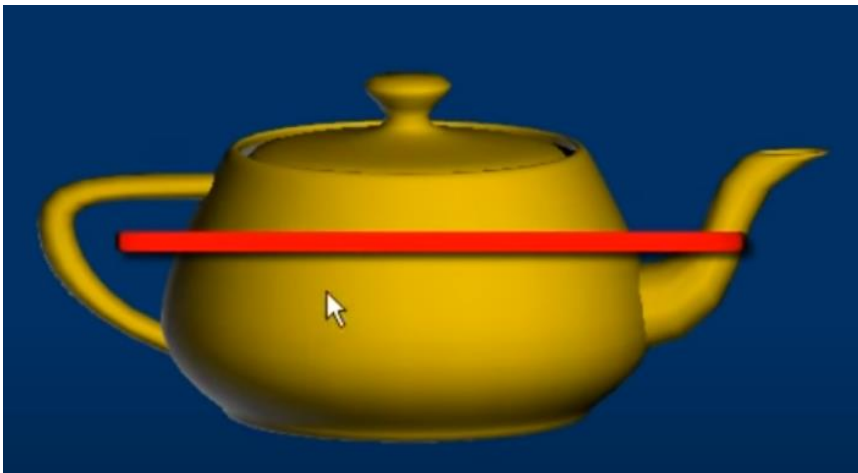
Most people will choose the top position

Why?: Because it provides the most visual information

How?: Rotate the tepot according to the PCA algorithm

Explained Visually

<https://setosa.io/ev/principal-component-analysis/>



Principal component Analysys a Layman's Introduction
<https://www.youtube.com/watch?v=BfTMmoDFXyE&t=30s>